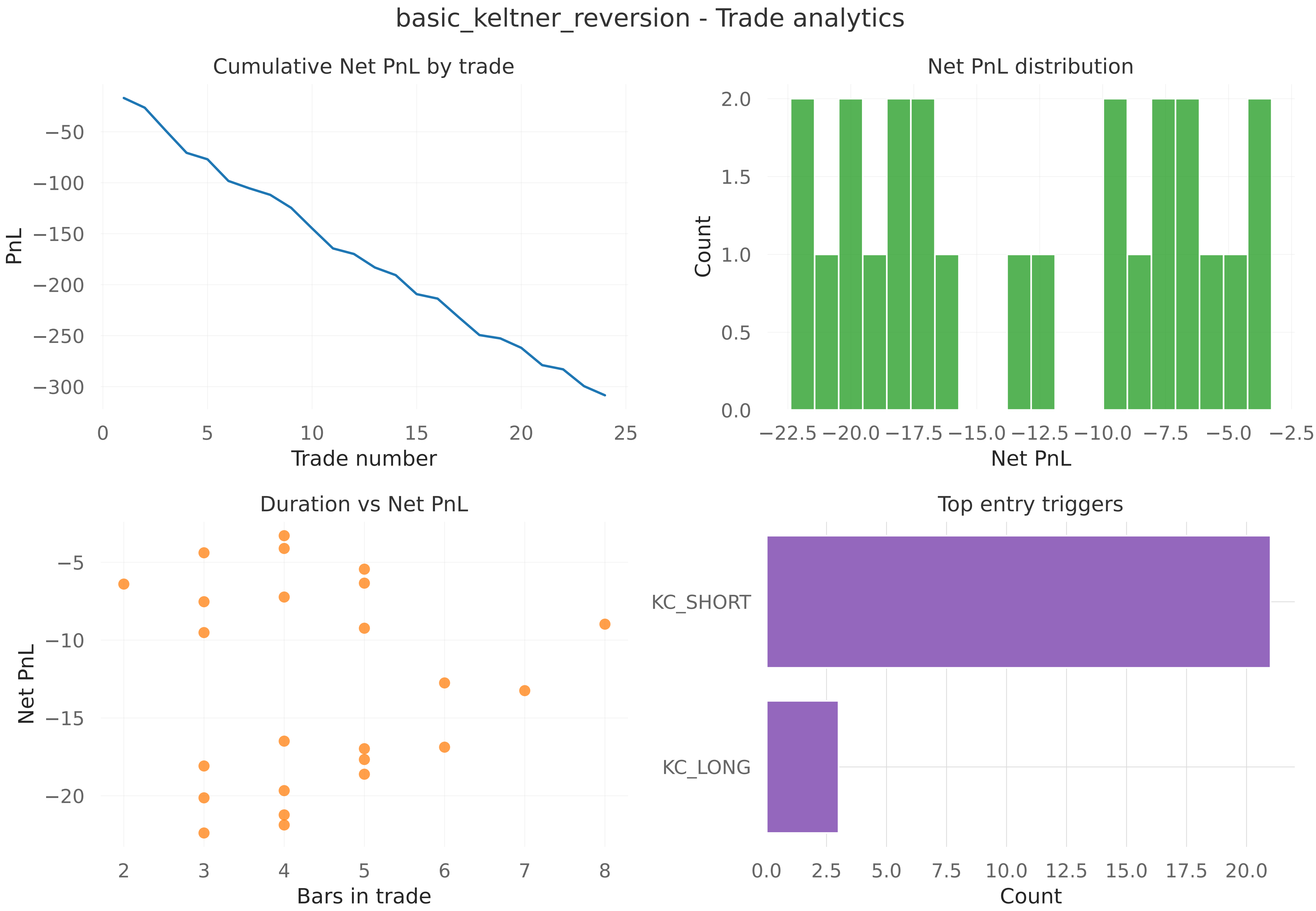
basic_keltner_reversion - Strategy Walkthrough

Strategy name basic_keltner_reversion Files reviewed -
src/strategies/basic_keltner_reversion/strategy.py -
src/strategies/basic_keltner_reversion/backtest_basic_keltner_reversion_v2.py Step-by-step e
flow 1) Backtest loads OHLCV data with fetch_ohlcv and returns early if data is empty. 2) A B
instance is created with fees, slippage, stop loss, take profit, pyramiding, and position siz
3) Loop starts after warmup = max(kc_ema_length, kc_atr_length) + 1. 4) For each bar, bt.on_b
first (intrabar stop/take-profit checks). 5) Strategy function keltner_reversion is called on
current bar. 6) keltner_reversion computes EMA and ATR, then Keltner upper/lower bands. 7) Si
to bt.on_signal with current close price and trigger text. 8) After loop, code builds stats,
image, optional QuantStats report, and optional trades CSV. Parameters and defaults - Strate
kc_ema_length=20, kc_atr_length=20, kc_atr_mult=1.5. - Execution and cost: initial_balance=10
commission_pct=0.001, slippage_pct=0.001. - Position controls: position_mode="all_in", trade_
position_pct=None, pyramiding=1, allow_short=True. - Risk controls: stop_loss_pct=0.02,
take_profit_pct=0.03. - Output flags: generate_report=True, generate_plots=True, generate_equ
Indicators/calculations - EMA(close, ema_length). - ATR(high, low, close, atr_length). - Kelt
upper = ema + atr * atr_mult, lower = ema - atr * atr_mult. Entry conditions - LONG when clo
Keltner band. - SHORT when close > upper Keltner band. Exit conditions - EXIT long trigger w
EMA. - EXIT short trigger when close <= EMA. Risk and position management
(SL/TP/ATR/trailing/pyramiding/position sizing) - Uses fixed percent stop loss and take profi
BacktesterV2 defaults in this runner. - No ATR-based stop in this strategy runner. - No trail
logic in strategy code. - Pyramiding is supported by BacktesterV2 (default 1). - Position siz
BacktesterV2 mode (all_in by default, optional position_pct). Outputs (trades, stats, plots,
clean_stats dictionary from bt.stats(). - Equity curve PNG under
static/backtests/basic_keltner_reversion/<run_id>/. - Trades CSV with trade details and metad
QuantStats HTML report when enabled. Known ambiguities or inconsistencies in the code - In s
the two EXIT checks cover all remaining prices, so function emits EXIT whenever price is betw
this can generate frequent EXIT signals even when no position exists. - export_trades_csv exp
named ema_fast and ema_slow although this strategy does not use EMA crossover params (metadat
to None).

basic_keltner_reversion - Trade summary

Trades: 24 Total Net PnL: -308.45 Average Net PnL per trade: -12.85 Win rate: 0.00% First tri
KC_LONG Last trigger: KC_EXIT_LONG Source CSV:
docs/strategy_walkthroughs/csv_baselines/basic_keltner_reversion.csv


basic_keltner_reversion - Trade analytics
Cumulative Net PnL by trade
PnL
−50
−100
−150
−200
−250
−300
0
5
10
15
20
25
Trade number
Net PnL distribution
Count
2.0
1.5
1.0
0.5
0.0
−22.5
−20.0
−17.5
−15.0
−12.5
−10.0
−7.5
−5.0
−2.5
Net PnL
Duration vs Net PnL
Net PnL
−5
−10
−15
−20
2
3
4
5
6
7
8
Bars in trade
Top entry triggers
KC_SHORT
KC_LONG
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Count